PHISHING DETECTION & AWARENESS REPORT

Project Title: Phishing Email Detection & Awareness System

Internship: Cybersecurity Internship

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1. Introduction

Phishing is a cyber-attack where attackers pose as a trusted source to steal sensitive data. It relies on **social engineering,** targeting the human firewall of an organization. Common techniques include impersonation, urgency manipulation, and fake verification requests.

- **Why it's dangerous:** It bypasses technical security filters by tricking the user into granting access willingly.

- **Awareness:** Training employees to spot these emails is the most effective way to prevent data breaches.

2. Email Overview

The following sample email was analyzed for this task:

- **Subject:** ⚠ Urgent: Your Account Will Be Locked

- **Sender:** Displayed as "Security Team" with no verifiable organizational domain, indicating potential sender spoofing.

- **Content Summary:** The email uses urgency-based language to pressure the recipient into clicking a verification link within 24 hours.

3. Identified Phishing Indicators

The following phishing indicators were identified during the security analysis:

1. **Generic Greeting:** "Dear User" is used instead of a specific name. Legitimate companies almost always use your registered name.
2. **Fear & Urgency:** The 24-hour deadline is designed to make the recipient panic and bypass critical thinking.
3. **Malicious Link:** The URL http://secure-account-verify-login.com is a **look-alike domain**. It uses keywords like "secure" to appear safe, but it is not an official company website.
4. **Lack of Branding:** There are no official logos, contact numbers, or specific account details provided.

4. Email Risk Classification

Risk Level: PHISHING CRITICAL

Reasoning: This email represents a classic credential harvesting attempt. The combination of psychological pressure tactics and a non-legitimate domain strongly indicates malicious intent.

5. How This Phishing Attack Works

1. **The Hook:** The attacker distributes the phishing email to multiple users to increase the probability of successful compromise.
2. **The Link:** A user clicks the link and is taken to a **fake login page** that looks identical to the real one.
3. **Credential Harvesting:** The user enters their credentials (username/password).
4. **The Result:** The attacker now has the credentials and can access the real account to steal data or money.

6. Technical Analysis

- **Domain Analysis:** The domain ***secure-account-verify-login.com*** is not associated with any legitimate organization and follows a suspicious naming pattern designed to mimic trusted services. The use of multiple security-related keywords increases deception likelihood.

- **Protocol:** The URL uses **HTTP instead of HTTPS**, indicating the absence of SSL/TLS encryption. Legitimate login portals use HTTPS to ensure secure data transmission.

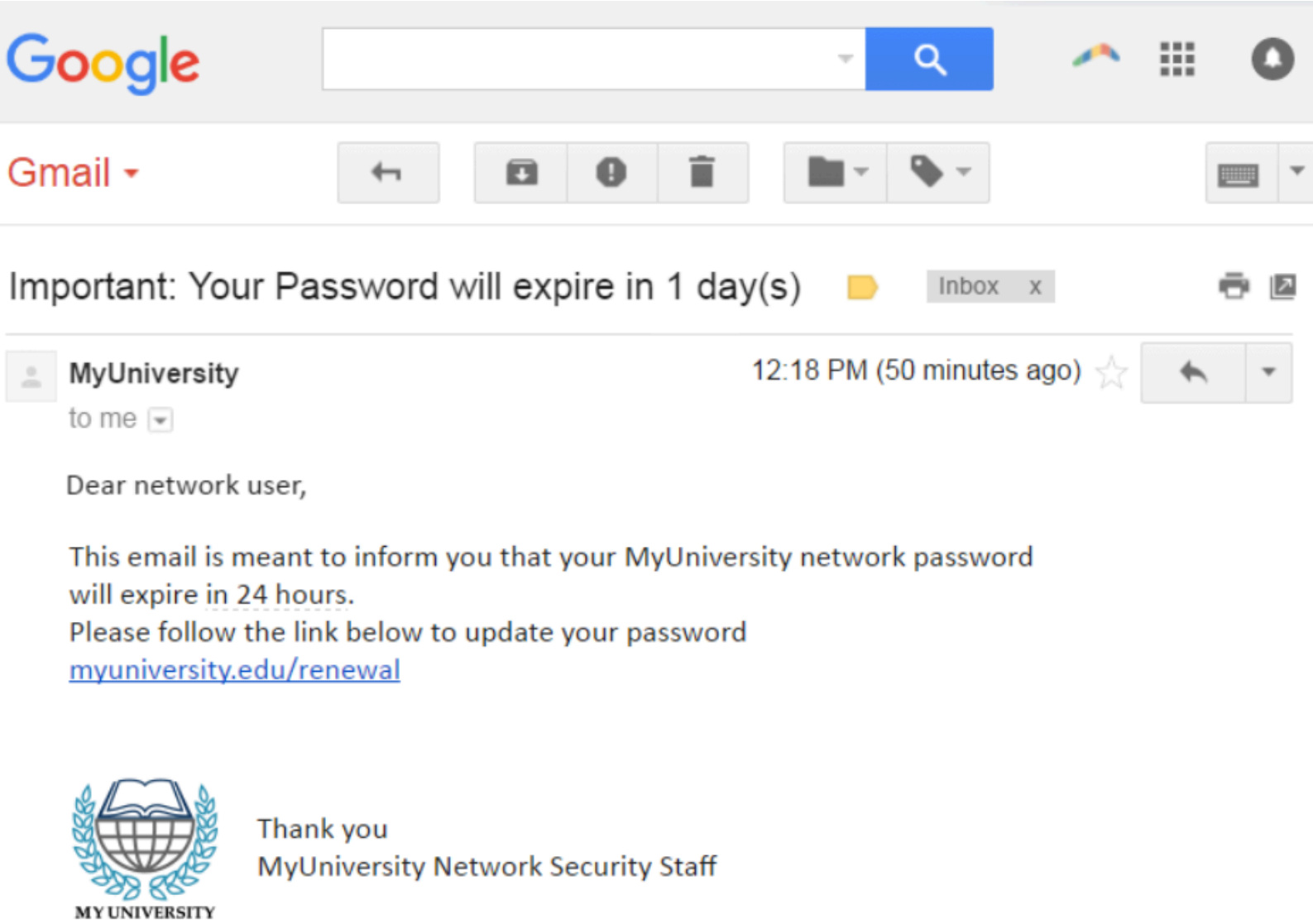
- **Fake Page:** The destination page is a spoofed login interface designed to capture user credentials and transmit them to the attacker-controlled server.

7. Prevention & Awareness Guidelines

Do’s and Don'ts for Employees:

- **DO** hover your mouse over links to see the real URL before clicking.
- **DO** report suspicious emails to the IT Security team.
- **DON'T** click links in emails that threaten to close your account.
- **DON'T** share passwords or MFA codes via email or unknown links.

8. Evidence – Sample Phishing Email Analysis



Email Summary

The email claims that the recipient's university network password will expire within 24 hours and urges immediate action by clicking a provided link to renew the password.

8.1 Identified Phishing Indicators

1. **Sense of Urgency:** The subject line states: *"Your Password will expire in 1 day(s)"* and the body says it will expire in 24 hours.
   This creates psychological pressure to act quickly without verifying authenticity.
2. **Generic Greeting:** The email begins with *"Dear network user"* instead of addressing the recipient by their real name.
   Legitimate institutions usually personalize official security emails.
3. **Suspicious Link:** The email provides a clickable link:
   myuniversity.edu/renewal

- Users should hover to check real URL
- Attackers often use look-alike domains

9. Conclusion

Awareness is the best defense. By verifying the sender and link before taking action, employees can stop a cyber-attack before it starts. **Always remember: Stop, Look, and Think before you click.**

10. References

- **NIST Cybersecurity Framework**
- **OWASP Phishing Guidelines**
- **Cybersecurity Awareness Training Standards**